

PRODUCT CIRCUIT DIAGRAM

A SYSTEMS VIEW OF HOW VALUE FLOWS THROUGH A PRODUCT — REVEALING FRICTION, MISMATCH, AND OPPORTUNITY

“Utilities are sort of impedance-matched with the government. If you look at a utility, they are heavily regulated by public utility commissions. So they’re sort of impedance-matched with the government—both literally and figuratively” —Elon Musk

🧠 CORE ANALOGY

Electrical Concept	Symbol	Product/system Equivalent
Voltage (V)	⚡⚡⚡	Strength of User Intent / Business Demand
Current (I)	—	Flow of Value / progress
Resistance (R)	—W—	Friction (UX, process, org, data)
Impedance / Mismatch	└┐	System Misalignment (handoffs, formats etc.)
Capacitor / Delay	— — —	Waiting / Buffering (queues, delays)
Amplifier	▶	Simplify / Automate / AI
Blackbox	■	Stable, trusted subsystem
Whitebox	□	Problem area to inspect
Measurement	🕒	Outcome (time, cost, success rate)

⚡ CREATING VALUE ⚡

Outcome Quality ∝ (Flow - Resistance - Mismatch - Delay) + Amplification

THE UNLOCK 🔒

Instead of asking:

“What feature should we build?”

Ask:

“Where is energy being lost in the system?”